安全警告

Security warning

1、本产品不能单独使用，必须在配备由本公司专业技术人员推荐的前级真空泵的前提下才可投入使用.

This product cannot be used alone. It must be used with a backing pump recommended by the company's professional technicians.

2、真空泵起吊时，可在泵所配的吊环螺钉处起吊，成套真空设备不得在泵所配的吊环螺钉处起吊。如果泵无起吊螺钉，可用铲车或其它搬运工具搬运真空泵。

When lifting the vacuum pump, use the lifting eye bolts provided. Do not lift the complete vacuum system using the pump's lifting eye bolts. If no lifting bolts are available, use a forklift or other handling tools.

3、 泵的电机连接架和电动机风扇防护罩均为半封闭结构，在泵运转时严禁用手或其它工具接触高速旋转的联轴器和电动机风扇叶片，以免对您造成不必要的伤害。

The motor connection frame and fan guard are semi - enclosed. Do not touch the high - speed rotating coupling or fan blades with hands or tools while the pump is running to avoid injury.

4、不得有任何限制真空泵排气的行为或在排气管路上安装任何截止阀门。真空泵也是一台压缩机，当泵在入口压力较高时，如果电机无过载保护装置，在真空泵的排气口会产生极高的压力，这种高压会引起真空设备的毁坏或对人员的伤害。

Do not restrict the pump's exhaust or install shut - off valves in the exhaust line. The vacuum pump is also a compressor. At high inlet pressure, without overload protection, extremely high pressure at the exhaust port may damage equipment or harm personnel.

5、真空泵系精密机械，在安装真空泵的进排气管道或阀门时严禁有任何异物进入泵腔内，以免引起不必要的严重后果。一旦有异物进入了泵腔，必须将泵腔内部的异物清除干净，否则严禁启动真空泵。如果系统配置的进排气管道的质量过重，必须对其进行支撑以免对真空泵精度的影响。

The pump is precision machinery. Prevent foreign objects from entering the pump cavity when installing pipes or valves. If foreign objects enter, remove them completely before starting. Support heavy pipes to avoid affecting pump precision.

6、泵不得用于抽吸含氧量高于 20% 的气体及其它易燃易爆气体。除非采用适当的防护装置及补充其它惰性气体，否则会引起爆炸并会毁坏真空泵和造成人员伤害

Do not pump gases with oxygen content > 20% or flammable/explosive gases. Without proper protection or inert gas supplementation, this may cause explosions, pump damage, or injury.

7、 在罗茨真空泵运行时严禁拆开油箱的加放、油螺塞进行加、放油，否则会对泵有严重的损害。

Do not open the oil tank plug to add/drain oil while the Roots pump is running, as this will severely damage the pump.

8、为防止前级真空泵排出的油雾及被抽气体长期对现场操作人员和环境的影响，不得将真空泵排出的气体直接排放在封闭的车间内或无良好通风性的建筑物内。

Do not discharge pump exhaust directly into enclosed workshops or poorly ventilated buildings to avoid long - term impact on operators and the environment.

9、在泵运行时，不得直接用手碰到灼热的真空泵表面。一般在低压工作时，泵的表面温度应不超 90℃。在高压运行时，泵的表面温度可能超过 100℃ 。

Do not touch the hot pump surface while it is running. Surface temperature should not exceed 90℃ at low pressure and may exceed 100℃ at high pressure.

10、在拆装或维护真空泵前务必将真空泵的电源切断，在确认真空泵已完全安装完毕后才允许接通电源。

Cut off the power before disassembling/maintaining the pump. Only connect power after confirming complete installation.

11、在维修真空泵时突然启动真空泵会造成设备的毁坏或对人员伤害。

Sudden startup during maintenance may damage equipment or harm personnel.

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I.概 述
Overview

1. 泵的结构特征与工作原理：

Structural Features and Working Principle of the Pump

ZJ(ZJB）型机械增压泵又称罗茨真空泵是一种容积式真空泵。它是利用一对8字型转子在泵壳中保持一定的间隙，作等速反向旋转而产生吸气和排气作用的（见图1)，二个转子被支承在泵壳的二侧面端盖的滚动轴承中，依靠一对可调节的同步齿轮使二个转子在高速旋转时始终保持着一定的相互位置，而转子与端盖之间的端面间隙是靠固定端轴承座下面的可调金属片来保证，使泵在工作时因受热而只能向一端膨涨。

The ZJ(ZJB) - type mechanical booster pump, also known as a Roots vacuum pump, is a positive - displacement vacuum pump. It relies on a pair of figure - 8 rotors that maintain a specific gap within the pump housing and rotate at a constant speed in opposite directions to achieve suction and exhaust (see Figure 1). The two rotors are supported by rolling bearings in the end covers on both sides of the pump housing. A pair of adjustable synchronous gears ensures that the two rotors always maintain a fixed relative position during high - speed rotation. The end - face gap between the rotors and the end covers is guaranteed by adjustable metal plates under the fixed - end bearing seat, allowing the pump to expand only in one direction when heated during operation.

端盖内的四组内轴封可防止两侧润滑油进入泵壳，出轴处油封可防止气体漏入泵内。

Four sets of internal shaft seals in the end covers prevent lubricating oil from both sides from entering the pump housing, while the oil seal at the shaft outlet prevents gas from leaking into the pump.

为减少泵温，除最小二种规格外，均有齿轮箱冷却器和出轴处油封冷却器。

To reduce the pump temperature, except for the two smallest specifications, all models are equipped with a gearbox cooler and a shaft - outlet oil seal cooler.

电动机的动力通过联轴器或三角胶带传给主动轴，再通过同步齿轮传动从动轴。

The power from the motor is transmitted to the driving shaft via a coupling or V - belt, and then to the driven shaft through the synchronous gear.

ZJ(ZJB）系列罗茨真空泵均制成卧式结构，中、小型泵采用联轴器与电机直联，大泵采用三角带传动。

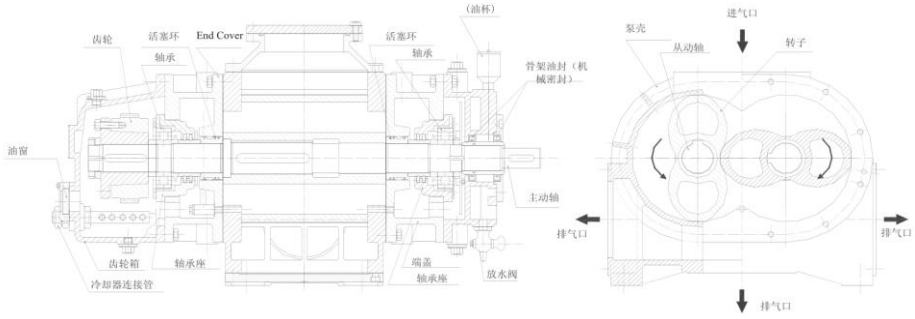
All ZJ(ZJB) series Roots vacuum pumps are of horizontal structure. Medium and small - sized pumps are directly connected to the motor using a coupling, while large pumps use V - belt transmission.

ZJB型带溢流阀罗茨真空泵，是ZJ型罗茨真空泵的派生产品，抽气原理与ZJ型泵一样，它也是利用一对8字形转子在泵壳中保持一定的间隙，作等速反向旋转而产生吸气和排气作用的。所不同的是ZJB型带溢流阀罗茨真空泵的进排气口相连通，在二者的通道上垂直地安装着一个自重阀（见图2）

The ZJB - type Roots vacuum pump with an overflow valve is a derivative product of the ZJ - type Roots vacuum pump. Its pumping principle is the same as that of the ZJ - type pump. It also uses a pair of figure - 8 rotors that maintain a certain gap in the pump housing and rotate at a constant speed in opposite directions to achieve suction and exhaust functions. The difference is that the inlet and outlet of the ZJB - type Roots vacuum pump with an overflow valve are connected, and a self - weight valve is vertically installed in the channel between them (see Figure 2).

其作用当进气口和排气口的压力差与阀的有效作用面积的乘积超过阀头的自重时，阀头就自动顶开。这个数值是设计的泵可靠运转的最高压力差值。因此这种阀实际上就是一种过载自动保护阀。这就是ZJB 型罗茨真空泵的最大优点。

Its function is that when the product of the pressure difference between the inlet and outlet and the effective acting area of the valve exceeds the self - weight of the valve head, the valve head will automatically open. This value is the maximum pressure difference for the designed pump to operate reliably. Therefore, this valve is actually an overload automatic protection valve. This is the greatest advantage of the ZJB - type Roots vacuum pump.



图/Figur 1:

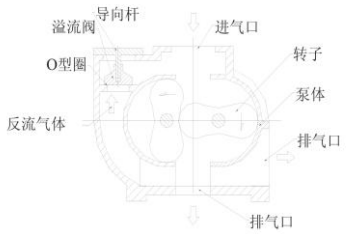
2．使用中的注意事项：

Precautions for Use

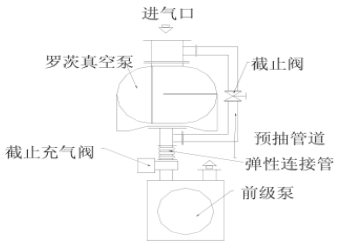
罗茨真空泵的最大优点是在较低入口压力时具有较高的抽气速率，但它不能单独使用，必须有一台前级真空泵串联，待被抽系统中的压力被前级真空泵抽到罗茨真空泵允许的起动入口压力时，罗茨真空泵才能开始工作（见图3)，并且在一般情况下，罗茨真空泵不允许在高压差时工作，否则将会使罗茨真空泵过载和过热而损坏，因此使用罗茨真空泵时必须合理地选用前级真空泵，并且安装必要的保护设备。

The greatest advantage of a Roots vacuum pump is its high pumping speed at low in let pressure. However, it cannot be used alone and must be connected in series with a backing pump. The Roots pump can only start working when the pressure in the system being pumped is reduced to the allowable starting inlet pressure of the Roots pump by the backing pump (see Figure 3). In general, the Roots pump is not allowed to operat

e under high pressure differentials, as this will cause overload and overheating, lea ding to damage. Therefore, when using a Roots pump, it is necessary to select the bac king pump reasonably and install necessary protective equipment.



图/Figur 2:



图/Figur 3:

前级真空泵一般为油封机械泵，但如果极限压力要求不高时，可选用其他形式的粗真空泵作为前级泵，特别在抽除含有大量水蒸汽的气体时，选用水环真空泵作为前级泵是很理想的，当然所能达到的极限压力要低些。

Backing pumps are generally oil-sealed mechanical pumps. However, if the ultimate pressure requirement is not high, other types of roughing pumps can be used as backi ng pumps. Especially when pumping gas containing a large amount of water vapor, using a water ring vacuum pump as a backing pump is ideal, although the achievable ultimat e pressure will be lower.

II. 主要技术参数
Main Technical parameters

1. ZJ型罗茨真空泵主要技术参数

Main Technical Parameters of ZJ-type Roots Vacuum Pump

型号/Model		ZJ-30	ZJ-70	ZJ-150A	ZJ-300	ZJ-600	ZJ-1200A
抽气速率（L/s）Pumping Speed		30	70	150	300	600	1200
极限压力（Pa）Ultimate Pressure		5×10^{-2}	5×10^{-2}	5×10^{-2}	5×10^{-2}	5×10^{-2}	5×10^{-2}
转速（r/min）Rotation Speed		2770	2870	2900	1450	2900	1450
口径(mm) Nominal Diameter	进气口 Lnlet Port	50	80	100	150	150	300
	排气口 Exhaust Port	40	50	100	150	150	300
重量/ Weight（Kg）		60	87	198	490	490	1550
配用功率 (kw) Motor Power		0.75	1.5	3	4	5.5	11
推荐选用前级真空泵 It is recommended to use a backing vacuum pump		2X-4C	2X-8	2X-15	2X-30A	2X-70A	X-150C

2.ZJB型罗茨真空泵主要技术参数

Main Technical Parameters of ZJB-type Roots Vacuum Pump

型号/Model		ZJB-30	ZJB-70	ZJB-150	ZJB-300	ZJB-600	ZJB-1200
抽气速率（L/s）Pumping Speed		30	70	150	300	600	1200
极限压力（Pa）Ultimate Pressure		5×10^{-2}	5×10^{-2}	5×10^{-2}	5×10^{-2}	5×10^{-2}	5×10^{-2}
启动压力（Pa）Starting Pressure		101326	101326	101326	101326	101326	101326
最大允许压差（Pa）Maximum allowable pressure difference		5332	5332	5332	4266	4266	4266
转速（r/min）Rotation Speed		2870	2870	2900	1450	2900	1450
口径.(mm）Nominal Diameter	进气口 Lnlet Port	50	80	100	150	200	300
	排气口 Exhaust Port	40	50	100	150	150	300
重量/Weight（kg）		75	100	215	480	503	1580
配用功率（kW）Motor Power		0.75	1.5	3	4	5.5	11
推荐选用前级真空泵 It is recommended to use a backing vacuum pump		2X-4C	2X-8	2X-15	2X-30A	2X-70A	X-150C

备注/Notes:

1，抽气速率系指在配用推荐前级泵的条件下，罗茨真空泵的入口压力在67 pa~2.67 pa 范围内所测得的最大抽气速率。

The pumping speed refers to the maximum pumping speed measured when the inlet pressure of the Roots vacuum pump is within the range of 67 Pa to 2.67 Pa, under the condition of using the recommended backing pump.

2，极限压力系指在配用推荐前级泵的条件下，不带任何附加容器、泵口密闭不进气，经过充分抽气后用压缩式真空计在泵进气口处测得的稳定的空气分压力的最低值。

The ultimate pressure refers to the minimum stable partial pressure of air measured at the pump inlet with a compression vacuum gauge after sufficient pumping, under the condition of using the recommended backing pump, without any additional containers and with the pump port sealed (no air intake).

3，上表各性能是在配用推荐前级泵的条件下获得的，用户可根据不同情况，选用不同的前级真空泵，但其主要性能指标会有所变化。

The performance data in the above table are obtained under the condition of using the recommended backing pump. Users can select different backing vacuum pumps according to different situations, but the main performance indicators will change accordingly.

III.安装说明

Installation Instructinons

安全说明/Safety Instructions:

罗茨泵的壳体及连接管道和附件表面的温度较高当人体接触时可能会造成伤害。

The surface temperature of the Roots pump casing, connecting pipes, and accessories is relatively high, which may cause injury if touched by the human body.

一旦操作不当，泵的内部或外部传动件及驱动设备也可能对人体造成严重损害。泵不得在拆开进气口盖或排气口盖的情况下运行。如果要对泵的运动部件进行检查或维护，务必首先完全切断和锁定控制电源设备。

Improper operation may also cause serious harm to the human body from the internal or external transmission parts and driving equipment of the pump. The pump must not be operated with the inlet or exhaust port cover removed. If inspecting or maintaining the moving parts of the pump, be sure to first completely cut off and lock the control power supply equipment.

在操作、起吊、安装、使用及维修设备时必须严格按照正确的处理程序进行。

When operating, lifting, installing, using, and maintaining the equipment, strictly follow the correct handling procedures.

1，安装场所：

Installation Site:

1）罗茨真空泵应安装在明亮少尘的建筑物内。

The Roots vacuum pump should be installed in a bright and dust-free building.

2）罗茨真空泵的安装环境温度为5~40C°。

The installation ambient temperature of the Roots vacuum pump is 5－40° C.

3）罗茨真空泵周围须留充分的余地，便于日常检查、维护保养、拆卸装配。

Sufficient space must be left around the Roots vacuum pump for daily inspection, maintenance, disassembly, and assembly.

2，基础工程：

Foundation Engineering:

1）罗茨泵可与前级泵等安装在同一机架上组成机组后再安装在混凝土基础上，大型泵必须单独安装，浇灌混凝土基础。

The Roots pump can be installed on the same frame as the backing pump to form a unit and then installed on a concrete foundation. Large pumps must be installed separately with a poured concrete foundation.

2）根据罗茨真空泵的重量和地基的耐压力确定混凝土基础的受压面积和深度。

Determine the pressure-bearing area and depth of the concrete foundation based on the weight of the Roots vacuum pump and the pressure resistance of the foundation.

3）为了在混凝土中埋进基础螺栓，混凝土基础浇灌时必须根据地基孔位置留出充分的空穴。

To embed foundation bolts in the concrete, sufficient cavities must be reserved according to the position of the foundation holes when pouring the concrete foundation.

4）基础表面做水平。

The foundation surface should be leveled.

3，安装：

Installation:

1）按底脚孔位置做好木框，将基础螺栓固定在木框上，放入预留孔中，再在预留孔中浇进混凝土，放置数日，待混凝土硬化后取掉木框。

Make a wooden frame according to the position of the foot holes, fix the foundation bolts on the wooden frame, place them in the reserved holes, then pour concrete into the reserved holes, and remove the wooden frame after the concrete hardens in a few days.

2）将罗茨真空泵放到基础上，用水平仪校水平，不平部分用金属垫片垫硬，旋紧基螺栓，用水泥将底面缝隙补满。

Place the Roots vacuum pump on the foundation, level it with a level, pad the uneven parts with metal shims, tighten the foundation bolts, and fill the bottom gaps with cement.

4，管道安装：

Pipeline Installation:

注意事项:在管道安装过程中如有焊渣及其它异物进入泵内会对罗茨泵和前级泵造成严重的损害毁坏，一旦出现类似情况务必对管道和泵进行严格清理。

Note :If welding slag or other foreign objects enter the pump during pipeline installation, it will cause severe damage to the Roots pump and the backing pump. In case of such a situation, the pipeline and pump must be strictly cleaned.

1）进出气管道内无任何焊渣、杂质等异物，有足够的通道能力。

Ensure no welding slag, impurities, or other foreign objects are present in the inlet and outlet pipes, and that the pipes have sufficient passage capacity.

2）罗茨真空泵出口至前级泵进气口的连接管道中须安装一段弹性软管，以减少前级泵的振动对罗茨真空泵的影响。

An elastic hose must be installed in the connecting pipe between the Roots vacuum pump outlet and the backing pump inlet to reduce the impact of backing pump vibration on the Roots vacuum pump.

3）为了缩短预抽时间，在一般的系统中，前级泵的进气口除与罗茨真空泵出口连接外，还可安装旁通预抽管道及阀门。

To shorten the pre-pumping time, in general systems, a bypass pre-pumping pipe and valve can be installed at the backing pump inlet in addition to the connection with the Roots vacuum pump outlet.

4）为了防止前级泵停止时的返油或返水（前级泵采用水环真空泵时），在前级泵的进口管道上应安装放气阀或逆止阀。

To prevent oil or water backflow (when a water ring vacuum pump is used as the backing pump) when the backing pump stops, an air release valve or check valve should be installed on the backing pump inlet pipe.

5）管道连接法兰处用O型橡胶密封圈密封，并涂上真空考克脂，整个真空系统须密封不漏。

Seal the pipe connection flanges with O-ring rubber gaskets and apply vacuum grease;

the entire vacuum system must be sealed without leaks.

6）泵的冷却水进口处管道上应装上阀门，以便调节水量。冷却水进出口的连接见第VIII部分的外形图。

A valve should be installed on the cooling water inlet pipe of the pump to adjust water flow. Refer to the outline drawing in Part VIII for cooling water inlet and out let connections.

5，电气设备安装：

Electrical Equipment Installation:

1）罗茨真空泵只能在规定的入口压力范围内工作，又当前级泵停止工作时，罗茨真空泵必须同时停止工作，因此，电器设备必须具有互锁及必要的自动保护。

The Roots vacuum pump can only operate within the specified inlet pressure range. When the backing pump stops, the Roots vacuum pump must stop simultaneously. Therefore, electrical equipment must have interlocking and necessary automatic protection.

2）电动机的接线必须使泵的转向符合箭头所指方向。

The motor wiring must ensure the pump rotates in the direction indicated by the arrow.

3）小泵可直接起动，大泵须装设起动器。

Small pumps can be started directly; large pumps require a starter.

IV.使用说明
Operation Instructions

1、起动前的准备工作：

1Pre-Start Preparation:

1）打开冷却水进水阀，检查冷却水是否畅通。

Open the cooling water inlet valve and check if the cooling water flows smoothly.

2）外伸轴处的油杯内注满润滑油，齿轮箱及轴承箱内的润滑油的油位须在油窗直径的3/4高度，润滑油用1号（粘度为46）真空泵油。

Fill the oil cup at the external shaft with lubricating oil. The oil level in the gearbox and bearing housing must be at 3/4 the height of the oil window. Use No. 1 (viscosity 46) vacuum pump oil.

3）凡在下列情况中使用时，必须采取相应的防护措施：

Corresponding protective measures must be taken when using the pump in the following situations:

a）吸入气体中有粉尘或颗粒时，必须在吸气口前安装除尘器或过滤器。

If the inhaled gas contains dust or particles, install a dust collector or filter b efore the suction port.

b）如果吸入气体中带有腐蚀性，必须采取中和措施。

If the inhaled gas is corrosive, take neutralization measures.

c）如果吸入气体中含由少量水蒸汽，所选用的前级泵又为油封机械泵时，为防止真空泵油的被乳化从而影响前级泵及机组的性能，油封机械泵应带有气镇装置，水蒸汽过多时，必须在机组进气口前或前级泵进气口前安装冷凝器。

If the inhaled gas contains a small amount of water vapor and the backing pump is an oil-sealed mechanical pump, the oil-sealed mechanical pump should be equipped with an gas ballast device to prevent emulsification of the vacuum pump oil (which affects the performance of the backing pump and unit). If there is excessive water vapor, install a condenser before the unit inlet or backing pump inlet.

2、起动程序：

Startup Procedure：

1）起动前级泵。

Start the backing pump.

2）打开前级泵预抽管道上的阀门和罗茨真空泵进排气阀门上。

Open the valve on the backing pump pre-pumping pipe and the inlet/outlet valves of the Roots vacuum pump.

3）待系统内的压力达到罗茨真空泵的允许入口压力后，将预抽管道上的阀门关闭，再起动罗茨真空泵。如无上述预抽管道则在前级泵抽到罗茨真空泵的允许入口压力后，即可起动罗茨真空泵。如采用自动控制装置，起动过程可实现自动方式。

After the pressure in the system reaches the allowable inlet pressure of the Roots vacuum pump, close the valve on the pre-pumping pipe, then start the Roots vacuum pump. If no such pre-pumping pipe is available, start the Roots vacuum pump after the backing pump has pumped down to the allowable inlet pressure of the Roots vacuum pump. If an automatic control device is used, the startup process can be automated.

3、运转中的注意点：

Operation Precautions：

1）运转中必须严格按照罗茨真空泵的技术规格使用，在正常情况下，罗茨真空泵的入口压力在1330pa以下，其最高入口压力应不超过它的允许最高入口压力。

Operate strictly in accordance with the technical specifications of the Roots vacuum pump. Under normal conditions, the inlet pressure of the Roots vacuum pump should be below 1330 Pa, and its maximum inlet pressure should not exceed its allowable maximum inlet pressure.

2）注意电动机负荷和泵的各部位温升情况，在正常运转情况下，泵的最高温升不得超过40C°，最高温度不得超过80C°。

Monitor the motor load and temperature rise of various parts of the pump. Under normal operation, the maximum temperature rise of the pump should not exceed 40° C, and the maximum temperature should not exceed 80° C.

3）运转中不应有不规则的异常振动和异常噪声。

There should be no irregular abnormal vibration or noise during operation.

4）如发现在运转中有电机过载，温升过高，声音异常，振动大等情况时，应立即停机检查原因，

排除故障。

If issues such as motor overload, excessive temperature rise, abnormal noise, or severe vibration are detected during operation, stop the pump immediately to check the cause and eliminate the fault.

4、停机程序：

Shutdown Procedure：

1）关闭前级油封机械泵气镇阀（如前级油封机械泵气镇阀在运转中开着气镇）。

Close the gas ballast valve of the backing oil-sealed mechanical pump (if it was open during operation).

2）关闭罗茨真空泵进气阀。

Close the inlet valve of the Roots vacuum pump.

3）先停罗茨真空泵，再停前级真空泵，严禁搞错停机程序。

Stop the Roots vacuum pump first, then the backing pump. Never reverse the shutdown sequence.

4）在停前级泵的同时，立即向前级泵进气口放大气。

When stopping the backing pump, immediately vent the inlet of the backing pump to the atmosphere.

5）关闭冷却水进水阀。

Close the cooling water inlet valve.

6）如需长时间停止使用，或在冬季为防止冻裂，必须将冷却水套内的剩水放尽。

If the pump will not be used for a long time, or to prevent freezing and cracking in winter, drain all residual water from the cooling water jacket.

V.维修与拆装
Maintenance and Disassembly

1、定期检查

Regular Inspection

1.1，每日检查：

Daily Inspection

（1) 油位检查：检查油杯、齿轮箱、轴承箱的油位，齿轮箱和轴承箱的油位过多，使温度升高，油位过低，造成润滑不良。

Oil Level Check: Inspect the oil level in the oil cup, gearbox, and bearing housing. Excessive oil in the gearbox and bearing housing will cause temperature rise, while insufficient oil will lead to poor lubrication.

（2) 温度检查：用温度计检查各部位温度。

Temperature Check: Use a thermometer to check the temperature of various parts.

（3）电动机负荷检查：用功率表或电流、电压表测量电动机负荷。

Motor Load Check: Measure the motor load using a power meter or current/voltage meter.

1.2，每月检查：联轴器弹性体或三角胶带的张力。

Monthly Inspection：Check the tension of the coupling elastomer or V-belt.

1.3，每季检查；齿轮箱内润滑油是否变质。

Quarterly Inspection：Check if the lubricating oil in the gearbox has deteriorated.

1.4，每半年检查：

Semi-Annual Inspection：

（1），前盖轴承箱内润滑油是否变质。

Check if the lubricating oil in the front cover bearing housing has deteriorated.

（2），油封是否损坏。

Check if the oil seal is damaged.

1.5，每一年检查：

Annual Inspection：

（1），轴承是否磨损。

Check if the bearings are worn.

（2），活塞环及活塞环衬套是否磨损。

Check if the piston rings and piston ring liners are worn.

（3），齿轮微量程度的磨损对转子正常工作是否产生影响是否需要调整。

Check if the slight wear of the gears affects the normal operation of the rotor and whether adjustment is needed.

2、拆装注意事项：

Precautions for Disassembly and Assembly：

罗茨真空泵进行拆装和重新装配时，须根据以下注意事项进行：

When disassembling and reassembling the Roots vacuum pump, the following precautions must be followed:

2.1，未拆卸前，先测量并记录转子各部分间隙。

Before disassembly, measure and record the clearances of all parts of the rotor.

2.2，尽量避免用重锤敲打，拆下的零件不得碰伤，妥善保管好。

Avoid heavy hammering as much as possible. Do not damage the removed parts and keep them properly.

2.3，将需要更换的零部件的更换原因及使用情况详细记录下来。

Record in detail the reasons for replacing the parts to be replaced and their usage.

2.4，重新装配前须把各另部件清洁干净，毛刺修光。

Before reassembly, clean all components thoroughly and smooth out any burrs.

2.5，无密封垫衬或密封圈的静密封面用“106”有机硅橡胶涂料密封，用于静密封的橡胶密封件需涂上真空考克脂。

For static sealing surfaces without gaskets or seals, use "106" silicone rubber coating for sealing. Rubber seals used for static sealing should be coated with vacuum cock grease.

2.6，根据间隙一览表调整转子各部间隙。

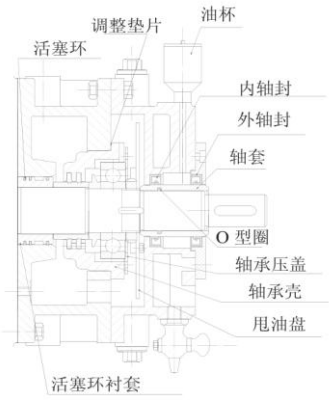
Adjust the clearances of all parts of the rotor according to the clearance list.

2.7，全部装好后须进行检漏。

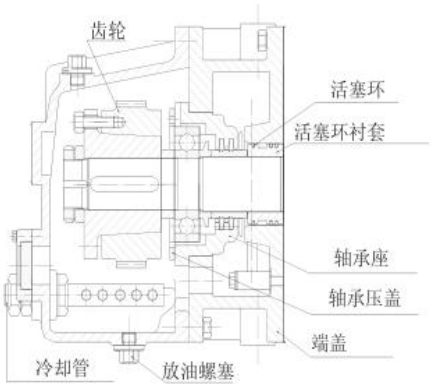
After full assembly, a leak test must be carried out.

2.8，重装后须进行试运转和必要的性能测试，待正常后才能安装使用。

After reassembly, a test run and necessary performance tests must be conducted. The pump can be installed and used only after it operates normally (see Figure 4 for the fixed end structure).



图/Figur 4:



图/Figur 5:

3、拆泵步骤：

Pump Disassembly Steps

拆泵前放尽泵内的积水和泵油并切断电源线

Drain all water and pump oil from the pump and cut off the power supply before disassembly.

3.1，拆固定端：（固定端结构见图4）

Disassemble the Fixed End: (See Figure 4 for the fixed end structure)

1）卸下电机及电机连接架；

Remove the motor and motor connection bracket.

2）拆下轴封压盖；

Remove the shaft seal gland.

3）拆下外轴封；（更换外轴封，在完成1~3步后，装上新轴封即可.）；

Remove the outer shaft seal. (To replace the outer shaft seal, simply install the new shaft seal after completing steps 1 - 3.)

4）拆下前盖；（更换内轴封，在完成1~4步后，装上新轴封即可.）；

Remove the front cover. (To replace the inner shaft seal, simply install the new shaft seal after completing steps 1 - 4.)

5）如须更换轴套，拉出轴套，用新的0型密封圈放入新轴套的密封槽后，把轴套压入泵轴即可；

If the shaft sleeve needs to be replaced, pull out the shaft sleeve, place a new O - ring into the sealing groove of the new shaft sleeve, and then press the shaft sleeve onto the pump shaft.

6）拆下轴承压盖；

Remove the bearing gland.

7）把轴承压盖和轴承从端盖拆出（更换轴承），拆下时注意将调整垫片妥善保管和安装的位置以便下次重新安装时使用；

Remove the bearing gland and bearing from the end cover (for bearing replacement). When removing, be careful to properly store the adjustment shims and note their installation positions for reuse during reinstallation.

8）拉出活塞环衬套和活塞环（更换活塞环或衬套）；

Pull out the piston ring liner and piston ring (for piston ring or liner replacement).

9）拆下端盖；

Remove the end cover.

10）压出保护套；

Press out the protective sleeve.

3.2拆活动端（活定端结构见图5）

Disassemble the Movable End (See Figure 5 for the Movable End Structure)

1）拆下后盖；

Remove the rear cover.

2）松开并拆下齿轮锁紧螺帽后拉出齿轮（分体齿轮须把齿圈和齿壳松开）；

Loosen and remove the gear lock nut, then pull out the gear (for split gears, loosen the gear ring and gear shell).

3）拆下轴承压盖；

Remove the bearing gland.

4）把轴承压盖和轴承从端盖拆（更换轴承）；

Remove the bearing gland and bearing from the end cover (for bearing replacement).

5）拉出活塞环衬套和活塞环（更换活塞环或衬套）；

Pull out the piston ring liner and piston ring (for piston ring or liner replacement).

6）拆下端盖；

Remove the end cover.

7）压出保护套；

Press out the protective sleeve.

8）拉出转子，拆泵完毕；

Pull out the rotor to complete the pump disassembly.

9）如果是带溢流阀罗茨真空泵须拆下溢流阀压盖，取出溢流阀阀头，检查导向套与阀头孔的上下运动是否灵活及密封圈是否可靠；

For Roots vacuum pumps with an overflow valve, remove the overflow valve gland, take out the overflow valve head, and check whether the up - and - down movement of the guide sleeve and valve head hole is flexible and whether the seal ring is reliable.

4、装泵步骤：

Pump Assembly Steps

装泵前须将各件清洁并修除毛刺并注意：各密封面不得上油！

Before assembly, clean all parts and remove burrs. Note: Do not apply oil to any sealing surfaces!

1）将一对转子放入泵体，注意长短轴转子的位置；

Place a pair of rotors into the pump body, paying attention to the positions of the long-shaft and short-shaft rotors.

2）在两端的端盖与泵体接触的平面上均匀涂上一层薄薄的106硅橡胶，紧固各螺钉；

Apply a thin, even layer of 106 silicone rubber to the contact surfaces between the end covers at both ends and the pump body, then tighten all screws.

3）装上活塞环及活塞环衬套和挡油套等部件；

Install components such as the piston ring, piston ring liner, and oil-retaining sleeve.

4）装上轴承座及轴承和轴承锁紧螺帽及轴承压盖（在固定端轴承座下将原拆下的调整垫片按原位置放入后，紧固轴承座）；

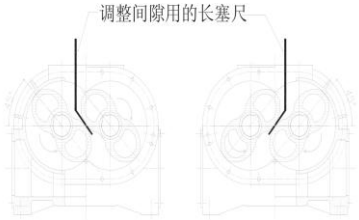
Install the bearing housing, bearing, bearing lock nut, and bearing gland. (For the fixed end, place the original adjustment shims in their original positions under the bearing housing before tightening the bearing housing.)

5）用塞尺测量转子与泵的两个端盖间的间隙（参数见间隙一览表）；

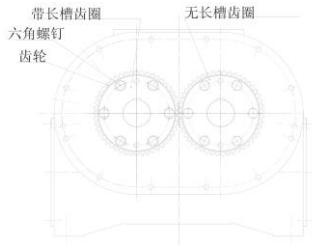
Use a feeler gauge to measure the clearance between the rotors and the two end covers of the pump (refer to the clearance list for parameters).

6） 在活动端装上齿轮及齿轮锁紧螺帽并略微拧紧锁紧螺帽后，（按图6，位置测量转子间隙）并根据间隙一览表参数调整两个转子间的间隙；

Install the gear and gear lock nut on the movable end, and after slightly tightening the lock nut, measure the rotor clearance according to the position shown in Figure 6 and adjust the clearance between the two rotors based on the parameters in the clearance list.



图/Figur 6:



图/Figur 7:

具体方法是：

Specific method:

首先将一个不带长槽的齿轮装入泵的主轴（见图7）并拧紧6个六角螺钉和圆螺母，然后将转子旋转至图6A位置。根据泵的转子与转子的间隙放入相应厚度的塞尺后，再将一个带长槽的齿轮装入从动轴，略紧六角螺钉。按照图6A和B位置重复测量转子间的间隙直到符合要求后，拧紧六角螺钉和圆螺母。

First, install a gear without a long slot onto the main shaft of the pump (see Figure 7), tighten the 6 hexagon screws and round nut, then rotate the rotor to position A in Figure 6. Insert a feeler gauge of the corresponding thickness according to the clearance between the pump rotors, then install a gear with a long slot onto the driven shaft and slightly tighten the hexagon screws. Repeat the measurement of the clearance between the rotors at positions A and B in Figure 6 until the requirements are met, then tighten the hexagon screws and round nut.

由于泵体与端盖的定位由圆柱销定位，因此，转子外圆与泵体内圆的间隙一般不需调整。

Since the pump body and end cover are positioned by cylindrical pins, the clearance between

the outer circle of the rotor and the inner circle of the pump body generally does not need to be adjusted.

间隙一览表

Clearance List

型号 Model	转子对泵体 Rotor to Pump Body	转子对转子 Rotor to Rotor	固定端 Fixed End	膨胀端 Expansion End
ZJ/ZJB-30	0.075-0.1	0.14-0.22	0.05-0.07	0.13-0.15
ZJ/ZJB-70	0.075-0.1	0.14-0.22	0.05-0.07	0.13-0.15
ZJ/ZJB-150A	0.12-0.14	0.22-0.26	0.09-0.11	0.15-0.18
ZJ/ZJB-300	0.19-0.22	0.24-0.32	0.12-0.15	0.24-0.28
ZJ/ZJB-600	0.19-0.22	0.24-0.32	0.12-0.15	0.24-0.28
ZJ/ZJB-1200A	0.20-0.25	0.60-0.70	0.17-0.20	0.35-0.38

7）在后盖平面上涂上106硅橡胶后，拧紧各螺钉。

Apply 106 silicone rubber to the rear cover surface, then tighten all screws.

8）在固定端装上挡油盘、轴套等零件，在前盖内装入内轴封后，在前盖平面上涂上106硅橡胶并紧固前盖；

Install parts such as the oil baffle disc and shaft sleeve on the fixed end. After installing the inner shaft seal in the front cover, apply 106 silicone rubber to the front cover surface and tighten the front cover.

9）带溢流阀罗茨真空泵须装上溢流阀阀头和压盖；

For Roots vacuum pumps with an overflow valve, install the overflow valve head and gland.

10）装上外轴封将轴封压盖后，装上电机连接架和电机，用手盘动联轴器应转动灵活，不得有卡住现象；

After installing the outer shaft seal and shaft seal gland, install the motor connection bracket and motor. Turn the coupling by hand; it should rotate smoothly without jamming.

11）如有可能对泵进行气密性试验，可在泵内打入2~3公斤/平方厘米的干燥空气，用肥皂液涂在各密封面，检查是否有漏气现象；

If possible, perform an airtightness test: Inject dry air at 2-3 kg/cm² into the pump, apply soapy water to all sealing surfaces, and check for leaks.

12）将各放油螺塞装上后给泵加油，注意油位高度一般为油窗中心；

Install all oil drain plugs, then fill the pump with oil. The oil level should generally be at the center of the oil sight glass.

13）接好电源线、水管及盖好泵的进气口盖；

Connect the power cord, water pipe, and cover the pump's inlet port.

14）启动泵，注意转向，如有必要检测泵的各项性能。

Start the pump, pay attention to the rotation direction, and test the pump's performance if necessary.

VI.故障原因及消除方法
Causes of Malfunction and Troubleshooting Methods

故障/Fault	原因/Cause	消除方法/Remedy
极限压力达不到 Insufficient Ultimate Pressure	1，管道、系统漏气 Leakage in pipes or system 2，泵部分漏气 Partial leakage in the pump 3，前级泵极限压力下降 Reduced ultimate pressure of the backing pump 4，润滑油太脏或牌号不对 Contaminated or incorrect type of lubricating oil 5，油封磨损 Worn oil seal 6，机封磨损 Worn mechanical seal	1，系统检漏 Leak test the system 2，对泵检漏 Leak test the pump 3，修理或更换前级泵 Repair or replacethe backing pump 4，调换润滑油 Replace the lubricating oil 5，调换油封 Replace the oil seal 6，调换机封 Replace the mechanical seal
抽速不足 Insufficient Pumping Speed	1，管道通道能力不够 Inadequate pipe passage capacity 2，前级泵抽速下降 Reduced pumping speed of the backing pump	1，增大管道通道能力 Increase pipe passage capacity 2，修理或更换前级泵 Repair or replace the backing pump
电动机过载 Motor Overload	1，入口压力过高 Excessively high inlet pressure 2，转子端面与端盖单面接触 Single-sided contact between rotor end face and end cover 3，前级泵返油进罗茨真空泵泵腔 Oil backflow from the backing pump into the Roots pump chamber	1，调整、控制入口压力 Adjust and control inlet pressure 2，调整转子端面间隙 Adjust rotor end face clearance 3，装置防返油设备 Install anti-backflow device
过热 Overheating	1，选择的前级泵抽速不够，造成压缩比过大 Insufficient pumping speed of the selected backing pump, causing excessive compression ratio 2，入口压力过高 Excessively high inlet pressure 3，冷却不良 Poor cooling 4，齿轮箱润滑油过多 Excessive lubricating oil in the gearbox 5，转子与泵壳接触 Rotor contact with pump casing 6，齿轮、轴承、油封润滑不良 Poor lubrication of gears, bearings, or oil seals	1，重新选用前级泵 Re-select the backing pump 2，调整、控制入口压力 Adjust and control inlet pressure 3，畅通、加大冷却水 Unblock or increase cooling water flow 4，调整油量 Adjust oil quantity 5，修整 Repair 6，保证油量适当，润滑良好Ensure proper oil quantity and good lubrication

声音异常 Abnormal Noise	1，装配不良 Poor assembly 2，导向齿轮与转子位置偏移，使转子相碰 Misalignment of guide gears and rotors, causing rotor collision 3，入口压力过高 Excessively high inlet pressure 4，过载或润滑不良造成对齿轮的损伤 Gear damage due to overload or poor lubrication 5，轴承磨损 Worn bearings	1，重装 Reassemble 2，调整位置、保证间隙 Adjust positions to ensure clearance 3，调整、控制入口压力 Adjust and control inlet pressure 4，调换齿轮 Replace gears 5，调换轴承 Replace bearings
轴承、齿轮早期磨损严重 Severe Early Wear of Bearings/Gears	1，润滑油不良 Poor lubricating oil 2，润滑油不足 Insufficient lubricating oil	1，调换润滑油 Replace the lubricating oil 2，补充润滑油 Add lubricating oil

VII.易预件一览表

Spare Parts List

型号/ Model	零件名称/ Part Name	型号/ Model/ Specification	数量/ Quantity
ZJ/ZJB-30	垫衬/ Gasket	/	1
	联轴器弹性体/ Coupling Elastomer	ML2	1
	骨架油封/ Skeleton Oil Seal	Φ30*Φ50*10	2
	机械密封/ Mechanical Seal	TF22	1
	活塞环/ Piston Ring	Φ38*Φ 34*2.5	8
	轴承/ Bearing	6205	4
	O型圈/ O-Ring	Φ15*Φ9*3	1
	O型圈/ O-Ring	Φ21*Φ 15*3	4
	O型圈/ O-Ring	Φ25*Φ20*2.5	1
	O型圈/ O-Ring	Φ63*Φ 55*4	1
ZJ/ZJB-70	垫衬/ Gasket	/	1
	联轴器弹性体/ Coupling Elastomer	ML2	1
	骨架油封/ Skeleton Oil Seal	Φ 30*Φ 50*10	2
	机械密封/ Mechanical Seal	TF22	1
	活塞环/ Piston Ring	Φ38*Φ34*2.5	8
	轴承/ Bearing	6205	4
	O型圈/ O-Ring	Φ15*Φ9*3	1
	O型圈/ O-Ring	Φ21*Φ15*3	4
	O型圈/ O-Ring	Φ25*Φ20*2.5	1
	O型圈/ O-Ring	Φ 92*Φ 84*4	1
ZJ/ZJB-150(A)	垫衬/Gasket	/	1
	联轴器弹性体/ Coupling Elastomer	ML3联轴器弹性体	1
	骨架油封/ Skeleton Oil Seal	Φ45*Φ62*12	2
	机械密封/ Mechanical Seal	TF30	1
	活塞环/ Piston Ring	Φ45*Φ41*2.5	12
	轴承/ Bearing	5207	4
	O型圈/ O-Ring	Φ15*Φ9*3	1
	O型圈/ O-Ring	Φ21*Φ15*3	1
	O型圈/ O-Ring	Φ 122*Φ 112*5	4
	O型圈/ O-Ring	Φ30*Φ22*4	4
	O型圈/ O-Ring	Φ38*Φ 30*4	1
	O型圈/ O-Ring	Φ49*Φ42*3.5	2

ZJ/ZJB-300/600	垫衬/Gasket	/	1
	联轴器弹性体/ Coupling Elastomer	ML5	1
	O型圈/ O-Ring	Φ21*Φ15*3	1
	活塞环/ Piston Ring	Φ 85*Φ78*2.5	12
	骨架油封/ Skeleton Oil Seal	Φ 65*Φ90*12(TCN)	2
	机械密封/ Mechanical Seal	TF50	1
	轴承/ Bearing	6311	4
	O型圈/ O-Ring	Φ 15*Φ9*3	1
	O型圈/ O-Ring	Φ 166*Φ154*6	4
	O型圈/ O-Ring	Φ 30*Φ22*4	4
	O型圈/ O-Ring	Φ 58*Φ50*4	1

ZJ/ZJB-1200(A)	联轴器弹性体/ Coupling Elastomer	ML7	1
	垫衬/Gasket	/	1
	活塞环/ Piston Ring	Φ115*Φ106*4	12
	骨架油封/ Skeleton Oil Seal	Φ120xΦ90x12	2
	机械密封/ Mechanical Seal	TF75	1
	轴承/ Bearing	N216W	2
	轴承/ Bearing	22216	2
	O型圈/ O-Ring	Φ 15*Φ9*3	1
	O型圈/ O-Ring	Φ366*Φ354*6	4
	O型圈/ O-Ring	Φ194*Φ180*7	1
	O型圈/ O-Ring	Φ85*Φ75*5	1
	O型圈/ O-Ring	Φ30*Φ22*4	6
	O型圈/ O-Ring	Φ21*Φ15*3	1

VII.外形尺寸和性能曲线
Overall Dimensions and Performance Curves

ZJ和ZJB型罗茨真空泵的外形尺寸均为采用Y型电机的实际寸。

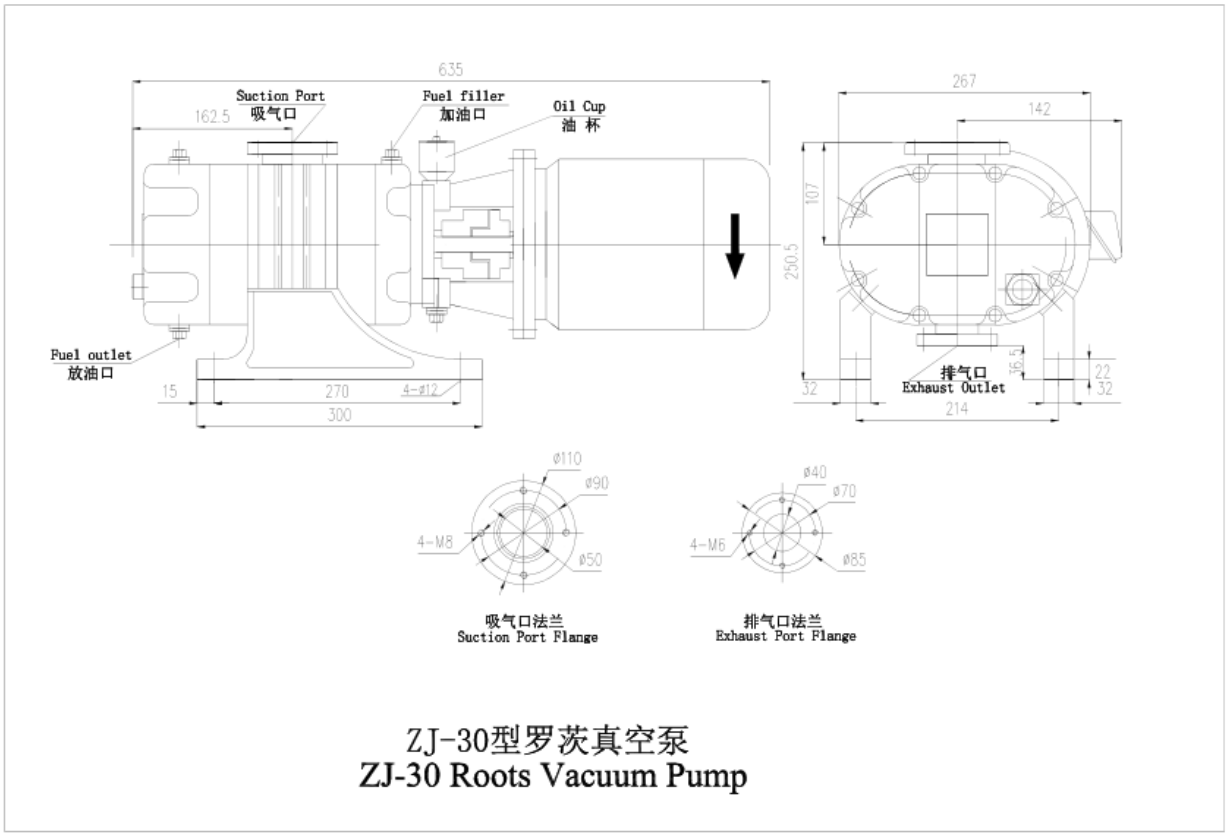
The overall dimensions of ZJ and ZJB(series Roots vacuum pumps are based on the actual dimensions when using Y-type motors.

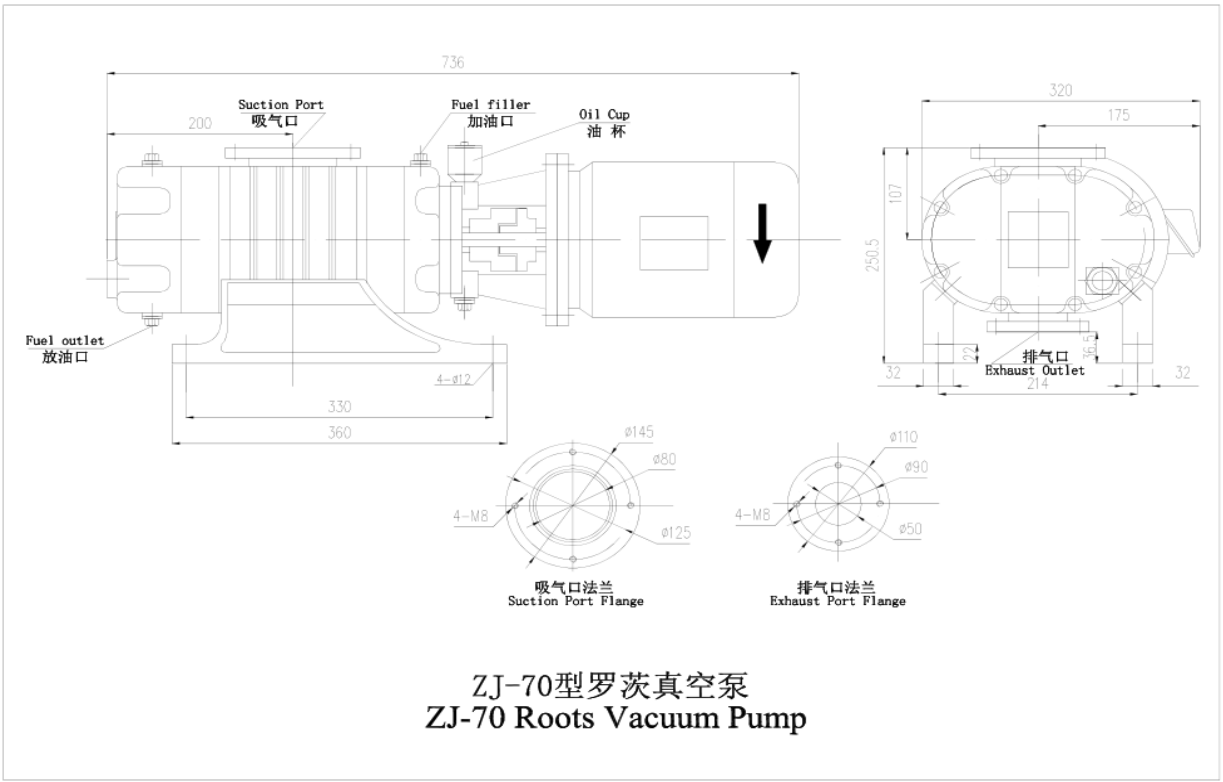
ZJ型和ZJB型罗茨真空泵的性能曲线是以推荐的前级泵时的实测线

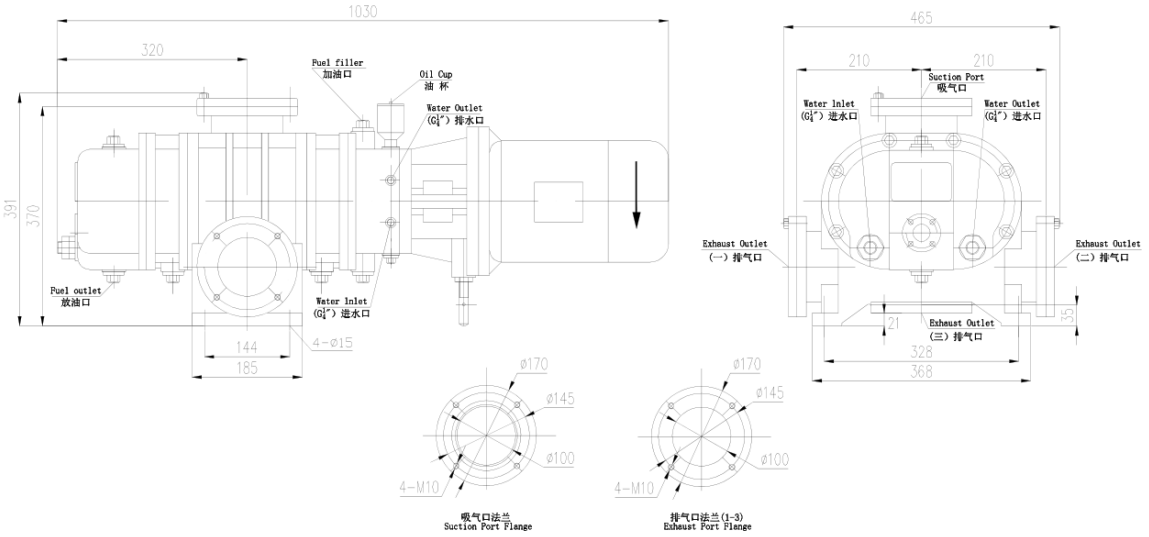
The performance curves of ZJ and ZJB series Roots vacuum pumps are measured with the recommended backing pumps.

1、ZJ-型罗茨真空泵外形图

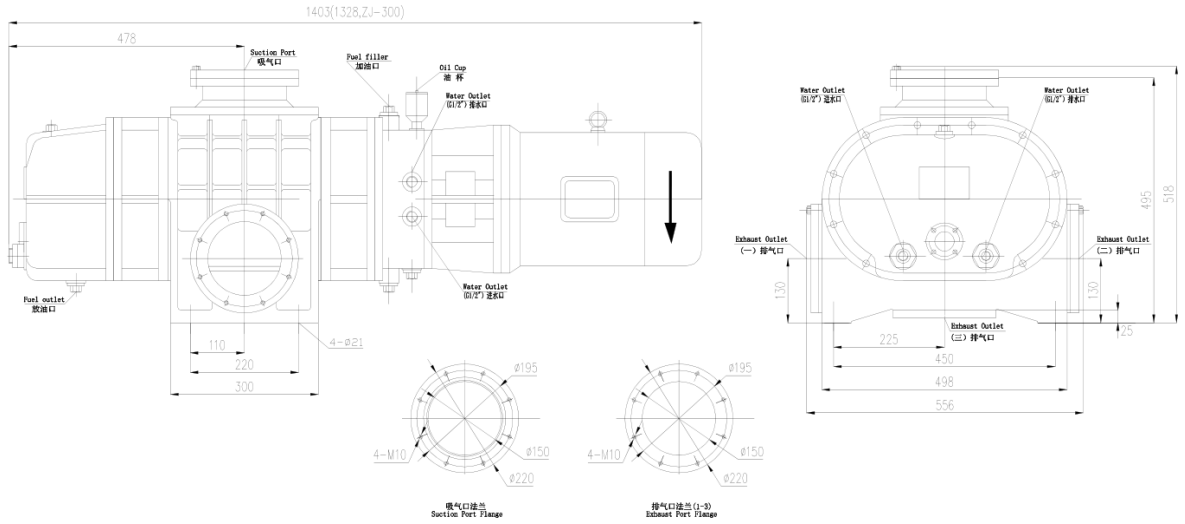
ZJ-type Roots Vacuum Pump Overall Dimensions



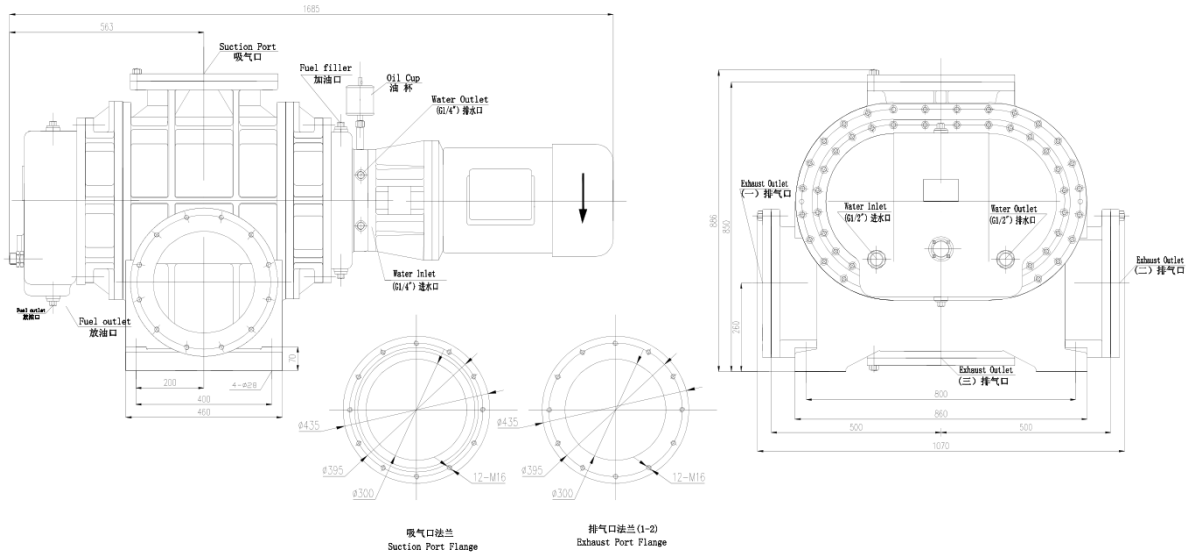




ZJ-150A型罗茨真空泵
ZJ-150A Roots Vacuum Pump



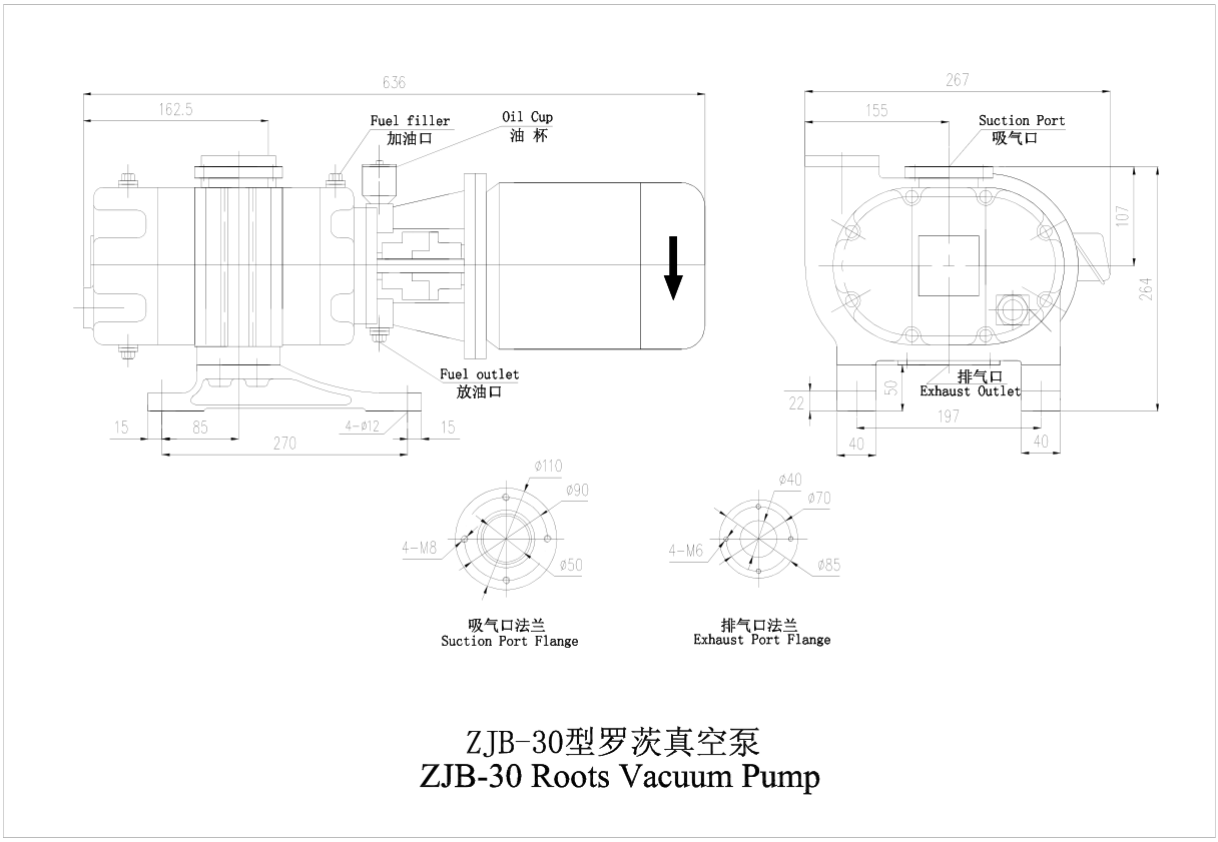
ZJ-300/600型罗茨真空泵
ZJ-300/600 Roots Vacuum Pump



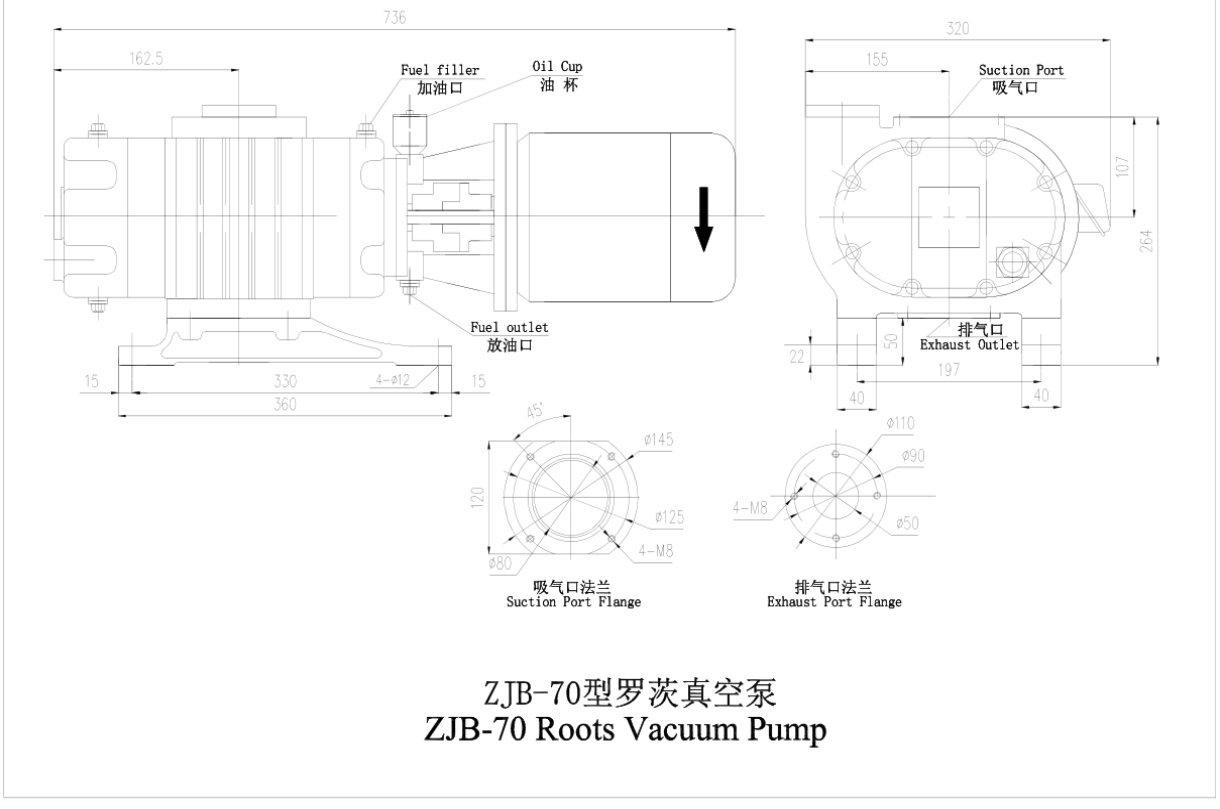
ZJ-1200A型罗茨真空泵
ZJ-1200A Roots Vacuum Pump

2、ZJB-型罗茨真空泵外形图

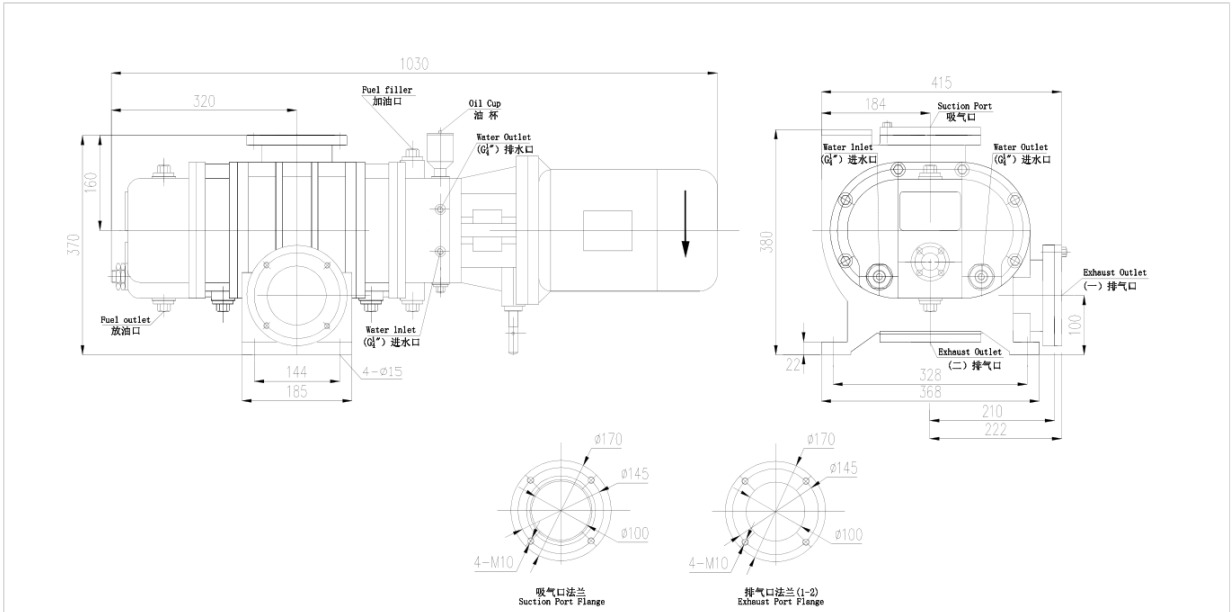
ZJB-type Roots Vacuum Pump Overall Dimensions



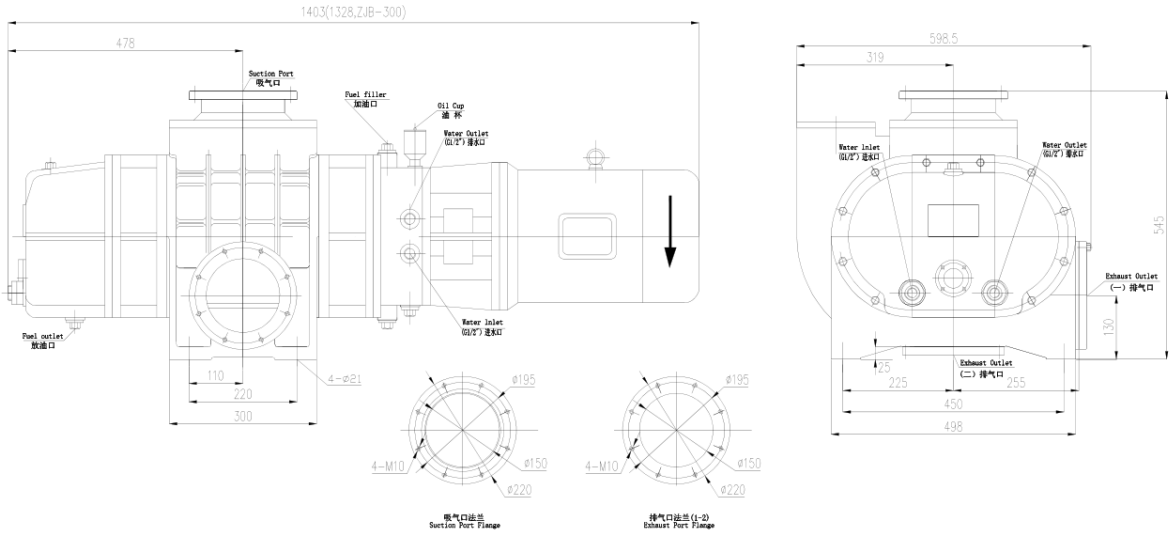
ZJB-30型罗茨真空泵
ZJB-30 Roots Vacuum Pump



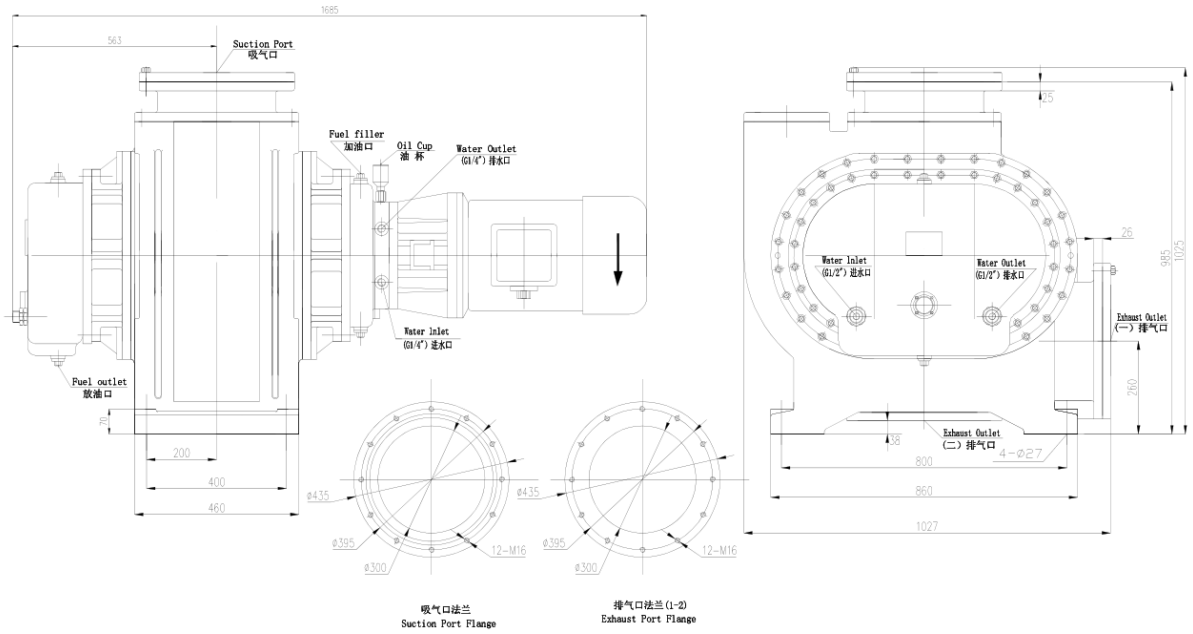
ZJB-70型罗茨真空泵
ZJB-70 Roots Vacuum Pump



ZJB-150型罗茨真空泵
ZJB-150 Roots Vacuum Pump



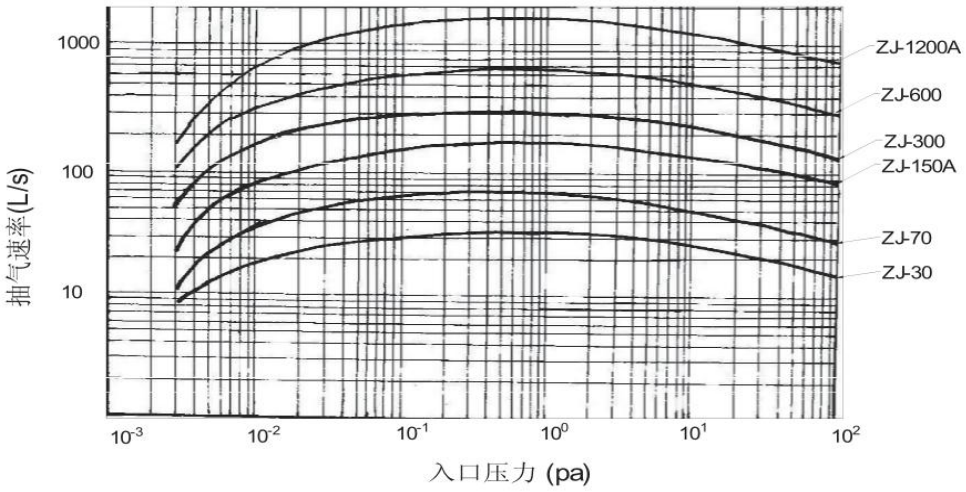
ZJB-300/600型罗茨真空泵
ZJB-300/600 Roots Vacuum Pump



ZJB-1200型罗茨真空泵
ZJB-1200 Roots Vacuum Pump

ZJ型罗茨真空泵性能曲线

Performance Curve of ZJ-type Roots Vacuum Pump



ZJB型罗茨真空泵性能曲线

Performance Curve of ZJB-type Roots Vacuum Pump

